WORK PROCEDURE **PRO-MEC-011A**

PROCEDURE FOR CHANGING THE COUPLING BETWEEN THE HYDRAULIC PUMP AND THE ELECTRIC MOTOR ON THE CUBEX DRILL

OBJECTIVE

The purpose of this procedure is to define a safe way of changing the coupling between the electric motor and the hydraulic pump on CUBEX drills. In addition, it allows a proper inspection of all components surrounding the pump and electric motor.

APPLICATION

This procedure applies to all MACHINES ROGER INTERNATIONAL employees who must perform coupling changes.
The work must be carried out by a mechanic, a supervisor or by a driller appointed by the supervisor.

INTERPRETATION

The person with authority to interpret this document is the Maintenance Manager. All requests for modifications, revisions or updates must be made to this person.

RESPONSABILITIES

It is the worker's responsibility to be aware of this procedure and to apply it when changing couplings.
The supervisor must ensure that the person performing the work is able to do so. The supervisor must ensure that the procedure is known to all concerned.

Tools and equipment required for work

Keys		**Other tools or products**
15/16	For the 2 bolts between the pump and the bell housing	**Antiseize or thread grease**
	For the 4 bolts between the bell housing and the electric motor (if hexagonals bolts installed)	**Pry bars**
ALLEN keys		**Steel brush**
5/16	For the lock bolt of the coupling on the pump	**Scraper**
3/16	For the lock bolt of the coupling on the electric motor	**Flat-blade screwdriver**
1/4	For the lock bolt of the spline of the coupling on the pump	**Hammer or sledgehammer**
1/2	For the 4 bolts between the bell housing and the electric motor (if ALLEN bolts installed)	**Rubber sledgehammer**

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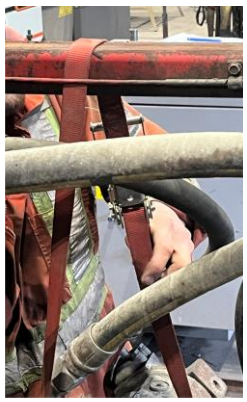
PROCEDURE

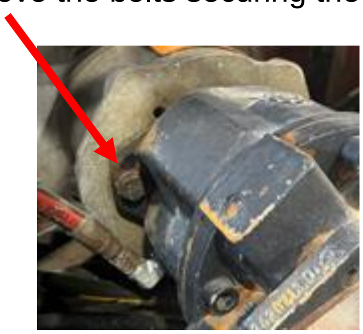
1- Apply the Lock out-tag out procedure;

2- Remove the hood on the top of the drill;

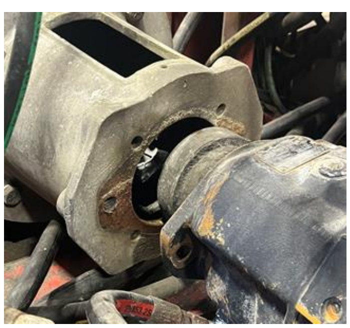
3- Wrap a ratchet strap around the pump and to the drill frame, not trapping the fire detection wire;



4- Remove the bolts securing the pump to the bell housing;



5- Remove the pump;

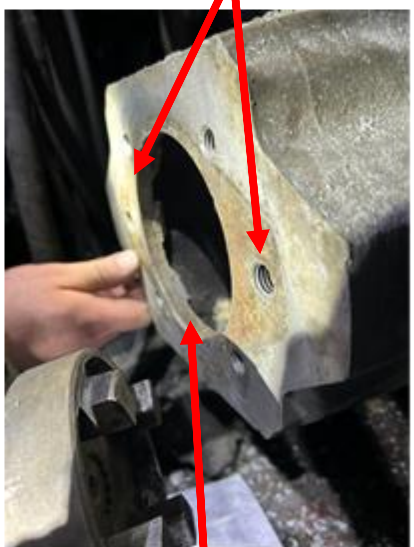


At this point, the pump is held in place by the hydraulic hoses and strap. It must be held in place with your hands on the hoses. AVOID GETTING YOUR HANDS UNDER THE PUMP, AND PROTECT YOUR FINGERS FROM GETTING TRAPPED.

6- Loosen the strap and position the pump to make sure it won't move in such a way as to hit the worker doing the work;

7- **Perform the inspection** :

a. Of the bell housing;
   i. Holes and threads where bolts are screwed



   ii. Hole of the bell housing,
   iii. Inside the bell housing

   iv. If there is dirt in the bell housing, remove the 4 bolts between the bell housing and the electric motor. Remove the bell housing and clean it with a scraper.

v. If the bell housing sticks to the electric motor, use a rubber hammer to tap it. **THE BELL HOUSING IS MADE OF ALUMINIUM**

8- Unscrew the pump coupling lock bolt using a 5/16 ALLEN key;

9- Remove the coupling from the pump shaft. Insert a pry bar into the slot in the couping to spread it apart, and using another pry bar, push the coupling to remove it towards

10-Remove the lock bolt on the coupling of the electric motor shaft using a 3/16 ALLEN key;

11- Remove the coupling using the pry bar;

IF THE COUPLING IS STUCK AND DIFFICULT TO REMOVE :

1- REMOVE THE BELL HOUSING
2- USE A SLEDGEHAMMER OR HAMMER TO STRIKE THE PRY BAR
3- BE CAREFUL NOT TO DAMAGE THE ELECTRIC MOTOR SHAFT

12-When both couplings are removed:

Using a steel brush or a scraper:

CLEAN

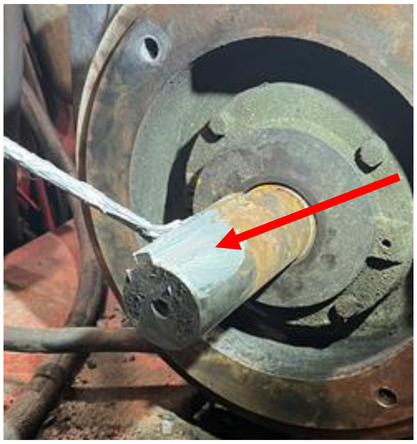
THE SHAFT AND THE JOINT BETWEEN THE PUMP AND THE BELL HOUSING

THE ELECTRIC MOTOR'S SHAFT

APPLY ANTSEIZE

ON THE SHAFT OF THE PUMP

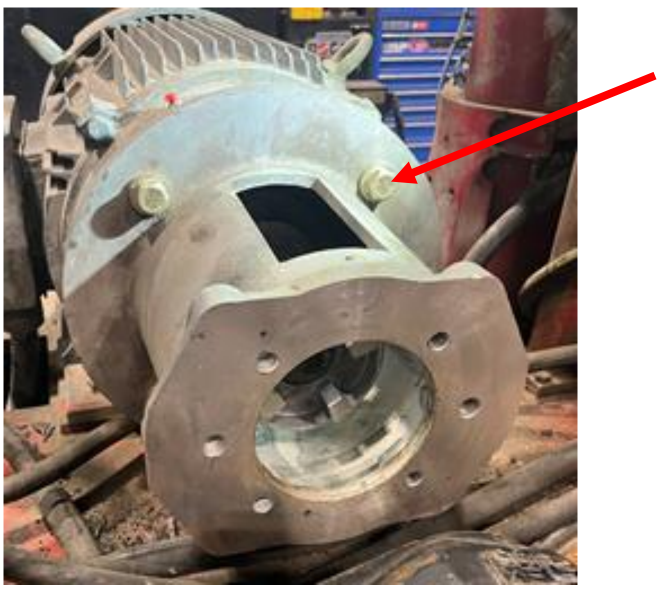
ON THE ELECTRIC MOTOR'S SHAFT



CLEAN BOTH ENDS OF THE BELL HOUSING

13-Insert the coupling floating on the electric motor shaft and tighten the lock by hand (just touching it). This will allow readjustment after entering the pump coupling

14-If the bell housing has been removed or replaced, reinstall it using the 15/16 wrench to tighten the 4 bolts;



15-Insert and tight the anti-seize coated bolt with the ¼" ALLEN wrench to lock the spline to the coupling.

16-Insert a small pry bar or a flat blade screwdriver into the slot on the top or front of the pump coupling to spread it apart and place the coupling on the shaft;

17-Make sure the coupling is "FLUSH" at the shaft;

18-Tighten the anti-seize coated lock bolt with the 5/16 ALLEN key;

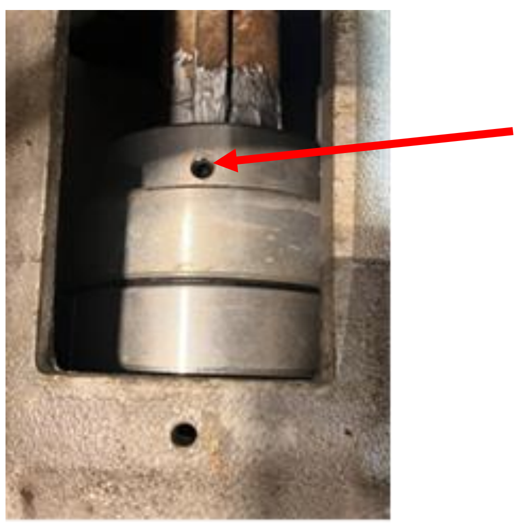
19-Insert the rubber gasket into the electric motor coupling;

20-Tighten the ratchet strap and align the two couplings together;

21-Insert the 2 hydraulic pump bolts by hand to ensure correct alignment;

22-Tighten the 2 bolts using 15/16 key;

23-Make sure the couplings are securely together and tighten the electric motor coupling lock with a 3/16 ALLEN key;



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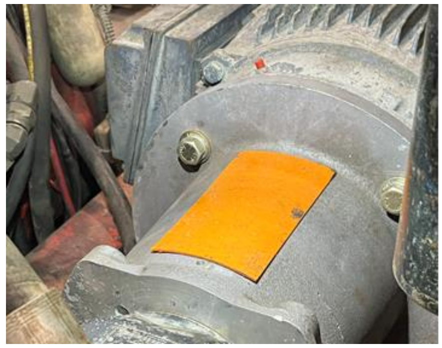
24-Replace the plastic cover



Karl Béchard
Maintenance Manager

Adds or modifications	Date	By
Procedure put in place after a fire broke out at the Westwood mine when the bell housing and plastic cover overheated.	April 2024	S. Tremblay K. Béchard